

Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

Among four men, P, Q, R and S, P takes thrice as much time as Q to complete a piece of work. Q takes thrice as much time as R, while R takes thrice as much time as S to complete the same work. One group of three of the four men can complete the work in 13 days, while another group of three can do so in 31 days. Which is the group that takes 13 days?

- ☐ a) **P, Q, S**
- ☐ b) **Q, R, S**
- ☐ c) **P, R, S**
- ☐ d) **P, Q, R**

Q2. DIRECTIONS *for questions 2 to 4:* Type in your answer in the input box provided below the question.

Find the maximum number of distinct regions into which the area inside a circle can be divided by drawing six chords.

Q3. DIRECTIONS *for questions 2 to 4:* Type in your answer in the input box provided below the question.

If $x + \frac{1}{x} = 1$, find the value of $x^{12} + x^9 + x^6 + x^3 + 1$.

Q4. DIRECTIONS *for questions 2 to 4:* Type in your answer in the input box provided below the question.

What is the last digit of $258^{35} + 393^{58}$?

Q5. DIRECTIONS *for questions 5 and 6:* Select the correct alternative from the given choices.

Two cars P and Q start from two points A and B towards each other simultaneously. They meet for the first time 40 km from B. After meeting they exchange their speeds as well as directions and proceed to their respective starting points. On reaching their starting points, they turn back with the same speeds and meet at a point 20 km from A. Find the distance between A and B.

☐ a) 130 km

☐ b) 100 km

☐ c) 120 km

☐ d) 110 km

Q6. DIRECTIONS for questions 5 and 6: Select the correct alternative from the given choices.

What is the sum of the series $3 + 33 + 333 + \dots$ to m terms?

☐ a) $\frac{8 \times 10^m + 10 - 9m}{27}$

☐ b) $\frac{8 \times 10^{m+1} + 27 - 9m}{27}$

☐ c) $\frac{8 \times 10^{m-1} - 27 - 9m}{27}$

☐ d) $\frac{10^{m+1} - 10 - 9m}{27}$

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

$f(x)$ is a quadratic function, such that $f(4) = 4 \times f(1)$. If one of the roots of $f(x) = 0$ is 2, find the second root.

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

Each of three friends, A, B and C, has a certain number of marbles with him, such that the number of marbles with B is 12% more than that with C and 12% less than that with A and C together. If the number of marbles with B is less than 500, then find the total number of marbles with all the three friends.

- ☐ a) **658**
- ☐ b) **940**
- ☐ c) **846**
- ☐ d) **Cannot be determined**

Q9. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.

A shopkeeper purchases ten watches at a price of ₹2000 each. He later finds out that two of the watches are defective and cannot be sold. What should be the selling price (in ₹) of each watch that he sells, if he is to earn a 15% profit overall?

Q10. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.

Two persons, A and B, simultaneously started running, from the same point, around a circular track, in the same direction, and met each other for the first time after 15 min from the start, during which time B had completed exactly five laps. Instead, if they had run in opposite directions, after how many minutes from the start would their 11th meeting have taken place, given that the speed of A is more than the speed of B?

Q11. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

Sumanta said to Tamal, "When I was half as old as you are today, you were one-sixth as old as I am now", If Sumanta is eight years older than Tamal, what is the sum of their present ages? (in years)

- ☐ a) 48
- ☒ b) 64 ✓ **Your answer is correct**
- ☐ c) 60
- ☐ d) 56

Q12. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

The provisions for a garrison of 240 men are sufficient to last for exactly 120 days. If after the first 40 days, a reinforcement of 60 men joins the garrison and every person decreases his daily consumption by 20%, for how many days more or less than the scheduled time will the provisions last?

- ☐ a) 10 days less
- ☐ b) 12 days more
- ☐ c) 40 days more
- ☐ d) None of the above

Q13. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

The amount of discount offered on an article was three times the profit made on selling it. If the article was marked up by 100% before offering the discount, then what was the percentage of profit made on selling it?

☐ a) 10%

☐ b) 20%

☐ c) 25%

☐ d) 75%

Q14. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

PQRS is a rectangle, of area 200 sq units, with length and breadth in the ratio of 2 : 1 (PQ > QR). PQT is a triangle in which PT = QT. If PT cuts SR at L and QT cuts SR at M and the area of triangle SLP is 30 square units, then the length (in units) of the altitude of the triangle PQT, through T, is

☐ a) $\frac{50}{3}$.

☐ b) 20.

☐ c) $\frac{70}{3}$.

☐ d) 25.

Q15. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

If the highest power of 225 in $n!$ is 10, find the number of values that n can assume.

- ☐ a) 0
- ☐ b) 3
- ☐ c) 5
- ☐ d) 10

Q16. DIRECTIONS *for question 11 to 16:* Select the correct alternative from the given choices.

Find the value of k for which the following pair of equations have an infinite number of solutions for (x, y) .

$$3x + 5y = 24$$

$$ky + 4x = 32$$

- ☐ a) 4
- ☐ b) $2\frac{2}{5}$
- ☐ c) $6\frac{2}{3}$
- ☐ d) 8

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

A cake is cut into three pieces, whose weights are in the ratio $1 : 2 : 3$. The heaviest piece is then further cut into four pieces, whose weights are in the ratio $1 : 2 : 3 : 4$. If at the end of this process, the lightest piece obtained weighs 24 gm, find the weight (in gm) of the original cake.

Q18. DIRECTIONS *for questions 18 to 22:* Select the correct alternative from the given choices.

The number of positive integers n , satisfying $n^3 - 16n - 4n^2 + 64 \leq 0$ is

- ☐ a) 2.
- ☐ b) 0.
- ☐ c) 1.
- ☐ d) More than 2.

Q19. DIRECTIONS *for questions 18 to 22:* Select the correct alternative from the given choices.

If $2a_n = a_{n-1} + a_{n+1}$, $a_8 = 46$ and $a_5 = 28$, then which of the following is true?

☐ a) $a_{11} = 62$

☐ b) $a_{10} = 58$

☐ c) $a_9 = 54$

☐ d) $a_7 = 42$

Q20. DIRECTIONS *for questions 18 to 22:* Select the correct alternative from the given choices.

Two circles, C_1 and C_2 , with respective radii of 13 cm and 17 cm and their centres 20 cm apart, intersect each other at two points P and Q. Find the area (in sq.cm) of the biggest triangle that can be drawn in the region where the two circles overlap, such that two of the vertices of the triangle are P and Q.

☐ a) $24\sqrt{30}$

☐ b) $8\sqrt{30}$

☐ c) $12\sqrt{30}$

☐ d) $6\sqrt{30}$

Q21. DIRECTIONS *for questions 18 to 22:* Select the correct alternative from the given choices.

A positive number N is divided into two unequal parts such that the difference of the cubes of the two parts is 133 times their difference. If the product of the two parts is 36, then what is the value of N ?

☐ a) 12

☐ b) 13

☐ c) 15

☐ d) 20

Q22. DIRECTIONS *for questions 18 to 22:* Select the correct alternative from the given choices.

Find the range of x , if $\frac{1}{x-2} > \frac{1}{2x}$.

- ☐ a) $(0, 2)$
- ☐ b) $(-1, 5)$
- ☐ c) $(-1, 0) \cup (3, \infty)$
- ☐ d) $(-2, 0) \cup (2, \infty)$